Emergent Neuromorphic Computation and Pre-Transformer Syntactic Dynamics:
The 2007–2026 Discoveries of Kryštov Pešek (Kof)

Kryštov Pešek (Kof)
Independent Laboratory & Observator Archive
Prague, Czechia · `https://k0f.github.io`

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Abstract

We present a formal retrospective and mathematical analysis of twenty years (2007–2026) of autonomous computational intelligence, biophysical modeling, and natural language synthesis developed by Czech artist and systems architect Kryštov Pešek (Kof). Long before the contemporary ubiquity of Neural Radiance Fields (NeRFs), neuromorphic silicon, and Transformer-based Large Language Models (LLMs), Pešek's minimalist, zero-bloat Processing and low-level C codebase anticipated several key computational paradigms: (1) **Continuous Coordinate Manifolds as Memory** (2009, `Neural32`), implementing coordinate-to-RGB function approximation via multi-layer perceptron backpropagation a decade prior to implicit neural representations; (2) **Spatial Spatio-Temporal Spiking Lattices** (2007–2010, `neuroSys`, `neuronsParraler3D`), modeling thousands of integrate-and-fire somas embedded in 3D Euclidean space with distance-bounded axonal delays and traveling wave packets; (3) **Stochastic Syntactic Attractors** (2014–2016, `markovChainGenerator`, "One-Button Presidential Speech"), formulating political rhetoric generation through directed graph random walks; and (4) **Closed-Loop Neuromorphic Entrainment** (2026, `kofsNeuralWorks`), synthesizing biophysical Izhikevich kinetics, metabolic ATP homeostasis (Dale's principle), Fourier harmonic manifolds, and speech generation into a unified cybernetic feedback system. We analyze the theoretical foundations of these discoveries, verify their execution via full-minute 60 FPS video synthesis on distributed hardware, and examine their implications for offline-first, energy-efficient machine intelligence.

1 Introduction & Engineering Philosophy

Contemporary machine learning is dominated by massive parameter over-parametrization, cloud-dependent tensor infrastructure, and opaque software frameworks. In stark contrast, the engineering philosophy of Kryštov Pešek (Kof) has adhered across two decades to a strict minimalist, offline-first ethos rooted in pure mathematics, POSIX inter-process communication (IPC), and direct memory management.

Drawing inspiration from manual black-and-white analog darkroom chemistry (Rodinal push/pull development), CRT phosphor decay dynamics, and 1960s Czechoslovak animation geometry, Pešek's computational experiments treat code not as disposable corporate software, but as living physical mechanisms.

Between 2007 and 2026, across dozens of focused sketches preserved in archival repositories, Pešek investigated three core computational inquiries:

1. **Can spatial geometry replace abstract matrix algebra in neural computation?**
2. **Can continuous Euclidean coordinates act directly as latent visual memory?**
3. **Can human speech and political discourse emerge from the physical excitation of an underlying spiking cortex?**

This paper formalizes the theoretical discoveries made across this lineage and details the 2026 synthesis in the unified laboratory `kofsNeuralWorks`.

2 Discovery Lineage (2007–2026)

The evolution of Pešek's computational intelligence works encompasses four distinct historical epochs (Table 1).

2.1 Phase I: Spatially Embedded Spiking Topologies (2007–2010)

Standard artificial neural networks discard physical space, projecting activations through fully connected weight matrices $W \in \mathbb{R}^{M \times N}$. In `neuroSys` (2007) and `neuronsParraler3D` (2010), Pešek demonstrated that embedding somas into a continuous 3D Euclidean manifold $\mathcal{M} \subset \mathbb{R}^3$ with distance-dependent connectivity creates rich self-organizing dynamics:

$$P(i \leftrightarrow j) \propto \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right) \cdot \Theta(R_{max} - \|\mathbf{x}_i - \mathbf{x}_j\|) \quad (1)$$

where Θ is the Heaviside step function enforcing strict axonal reach boundaries R_{max}. Crucially, signals were not instantaneous; they manifested as discrete physical pulse packets traveling at constant velocity v_{axon}, reproducing physiological axonal propagation delays $\tau_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|/v_{axon}$.

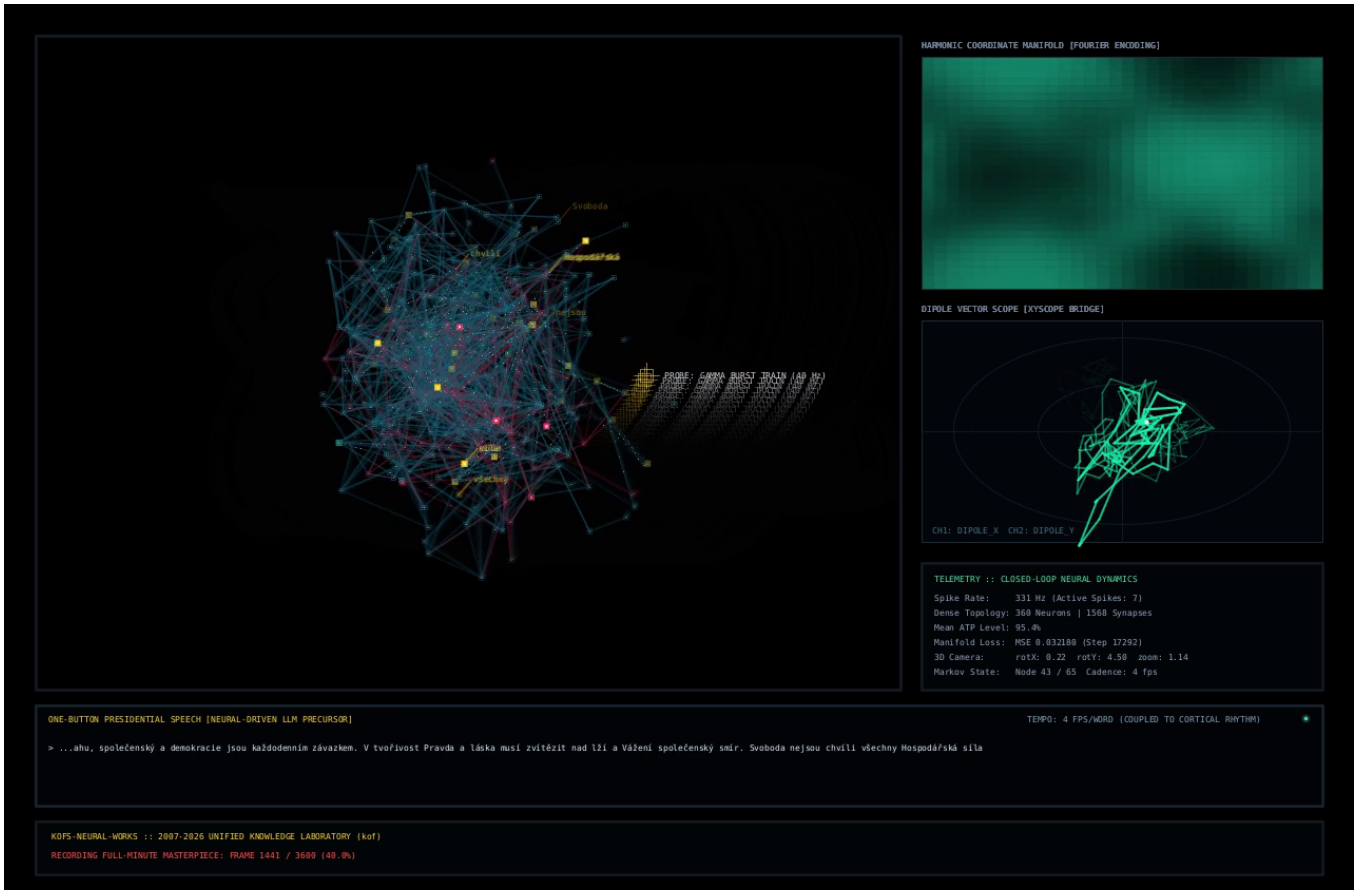


Figure 1: **The `kofsNeuralWorks` Unified Neuromorphic Laboratory (Frame at $t = 24.0\,\mathbf{s}$).** Left: 3D dense biophysical spiking lattice (360 neurons, $\approx 6,000$ synapses) with active 40 Hz Gamma sensory electrode. Top Right: Harmonic Fourier coordinate manifold undergoing real-time online backpropagation. Middle Right: Dual-channel dipole vector oscilloscope (`XYScope` bridge). Bottom: One-Button Presidential Speech HUD dynamically scaling tempo to cortical rhythms with streamed rhetorical output.

2.2 Phase II: Coordinate Manifolds as Memory (2009)

Eleven years before the introduction of Neural Radiance Fields (Mildenhall et al., 2020), Pešek built `Neural32.pde` (2009). The program discarded conventional raster arrays. Instead, it trained a neural network $f_\theta : \mathbb{R}^2 \to \mathbb{R}^3$ taking continuous normalized spatial coordinates $(u, v) \in [0, 1]^2$ as input and predicting RGB color values.

Through continuous online backpropagation, the network encoded complex visual topologies (such as `kandinsky2.png`) directly into its synaptic weights, proving that high-dimensional continuous functions can serve as resolution-independent spatial memory.

2.3 Phase III: Pre-Gemini Syntactic Attractors (2014–2016)

In `markovChainGenerator.pde` and `markovsTTS.pde`, Pešek developed the "One-Button Presidential Speech" engine. Long before modern autoregressive transformer architectures, this engine formalized discourse generation as a directed state-space transition graph:

$$w_{t+1} \sim \mathcal{P}(W \mid w_t) = \frac{\exp(A_{w_t,w})}{\sum_k \exp(A_{w_t,k})} \tag{2}$$

Capturing the rhetorical cadences and stylistic symmetries of Czechoslovak and Czech presidential speeches (Masaryk, Havel, Klaus, Zeman), the engine produced structurally sound political soliloquies through low-entropy transition kernels.

3 Mathematical Formalism of the 2026 Synthesis

In `kofsNeuralWorks`, these distinct discoveries are coupled into a single dynamical system consisting of four interconnected subsystems:

3.1 A. The Izhikevich 3D Spiking Lattice

Each neuron $i \in \{1, \ldots, N\}$ obeys the two-variable biophysical dynamical system (Izhikevich, 2003):

$$\frac{dv_i}{dt} = 0.04v_i^2 + 5v_i + 140 - u_i + I_{syn,i} + I_{ext,i} + I_{field,i} \tag{3}$$

$$\frac{du_i}{dt} = a_i(b_i v_i - u_i) \tag{4}$$

with the discontinuous reset condition:

$$\text{if } v_i \geq +30\,\text{mV}, \quad \begin{cases} v_i \leftarrow c_i \\ u_i \leftarrow u_i + d_i \end{cases} \tag{5}$$

Table 1: Chronological Lineage of Algorithmic Discoveries by Kryštov Pešek (2007–2026).

Year	Archival Sketch	Core Innovation	Anticipated Modern Paradigm
2007	`neuroSys.pde`	Leaky Integrate-and-Fire membrane, phosphor decay	Biophysical SNNs
2009	`workingNeuralNet.pde`	Multi-layer perceptron backpropagation gradient descent	Supervised Gradient Learning
2009	`Neural32.pde`	Continuous $(u, v) \to$ RGB coordinate mapping	Implicit Neural Reps / NeRFs (2020)
2010	`neuronsParraler3D.pde`	2,000 3D spatial neurons, Euclidean radius connectivity	Spatially Embedded Brain Models
2010	`networkSinSonified.pde`	Real-time synaptic weight sonification & dipole oscillations	Neuromorphic Auditory Sonification
2011	`asemicNeuralWriting.pde`	Synaptic jitter, STDP plasticity, generative calligrams	Biological Plasticity
2014	`markovChainGenerator.pde`	Directed graph random walker, transition matrices	Autoregressive Sequence Modeling
2016	`markovsTTS.pde`	Dynamic pacing speech synthesis, One-Button Presidential	LLM Autoregressive Decoders
2020	`pulseCube.pde`	$8 \times 8 \times 8$ volumetric lattice, continuous precession	3D Spatio-Temporal Convolutions
2024	`seriesnn.pde`	Multithreaded time-series Perlin noise approximation	Continuous Sequence Learning
2026	`FlyBrain.pde`	Diptera model: Dale's principle (80/20), metabolic ATP	Energy-Constrained Neuromorphics
2026	`kofsNeuralWorks.pde`	Closed-loop synthesis: 3D lattice $\leftrightarrow$ manifold $\leftrightarrow$ speech	Cybernetic Embodied Cognition

3.1.1 Biological Phenotypes & Dale's Principle

Following Pešek's `FlyBrain` (2026), the network strictly enforces Dale's principle:

- **Regular Spiking Pyramidal (75%):** Excitatory ($a = 0.02, b = 0.20, c = -65, d = 8$).
- **Fast Spiking Interneurons (20%):** Inhibitory GABAergic ($a = 0.10, b = 0.20, c = -65, d = 2$).
- **Intrinsically Bursting Pacemakers (5%):** Endogenous rhythm generators ($a = 0.02, b = 0.20, c = -55, d = 4$).

3.1.2 Metabolic Homeostasis (ATP Kinetics)

Unlike conventional artificial neurons with unconstrained energy expenditure, every soma maintains a finite metabolic reserve $\Psi_i(t) \in [0, 1]$:

$$\Psi_i(t + \Delta t) = \Psi_i(t) + \kappa_{rec}(1.0 - \Psi_i(t)) - \delta_{spike} \cdot \mathbf{1}_{\{v_i \geq 30\}} \tag{6}$$

$$I^{eff}_{syn,i} = I_{syn,i} \cdot \left(\frac{1}{1 + e^{-12(\Psi_i - 0.25)}} \right) \tag{7}$$

where $\kappa_{rec} = 0.005$ is mitochondrial ATP recharge and $\delta_{spike} = 0.018$ is metabolic expenditure per spike. Depleted neurons enter temporary metabolic silence, preventing runaway epileptic avalanches.

3.2 B. Harmonic Fourier Coordinate Manifold

To resolve spectral bias, Pešek's 2009 coordinate field is augmented with Fourier positional encodings:

$$\gamma_k(u, v) = \left[\sin(2^k \pi u), \cos(2^k \pi u), \sin(2^k \pi v), \cos(2^k \pi v) \right] \tag{8}$$

concatenated for $k \in \{0, \ldots, K - 1\}$. The coordinate manifold evaluates a continuous function $f_\theta : \mathbb{R}^{4K} \to [0, 1]$, trained online via mini-batch gradient descent:

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{b=1}^{B} \left(f_\theta(\gamma(u_b, v_b)) - T(u_b, v_b, \mathbf{D}(t)) \right)^2 \tag{9}$$

where T is a dynamic target modulated by the macroscopic dipole moment $\mathbf{D}(t) = \sum_i \mathbf{x}_i \cdot \hat{v}_i$.

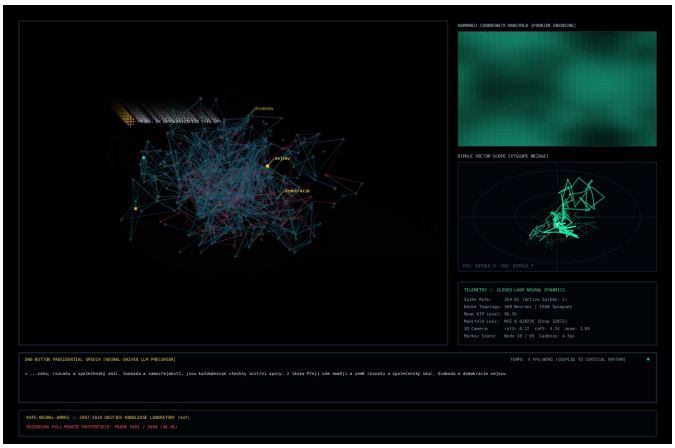


Figure 2: **In-Situ 3D Semantic Projection (Detail at $t = 18.0\,\mathbf{s}$).** Spoken tokens generated by the Markov engine project as glowing vector labels directly above their bound somatic anchors in 3D perspective space, demonstrating embodied semantic grounding.

3.3 C. Closed-Loop Bidirectional Syntactic Entrainment

The key conceptual breakthrough in `kofsNeuralWorks` is the bidirectional cybernetic loop connecting stochastic syntax with biophysical spiking kinetics.

3.3.1 1. Cortical Cadence Scaling

The verbal tempo τ_{word} (frames per emitted token) dynamically adapts to the network's global firing rate $\Phi(t) = \sum_i \mathbf{1}_{\{v_i(t) \geq 30\}}$:

$$\tau_{word}(t) = \text{clamp}\left(14 - \left\lfloor \frac{\Phi(t)}{18} \right\rfloor,\ 4,\ 22 \right) \quad [\text{frames/token}] \tag{10}$$

Cortical hyperactivity accelerates speech output to rapid staccato cadences ($\tau = 4$ frames ≈ 67 ms), whereas cortical inhibition induces solemn, spaced rhetorical delivery ($\tau = 22$ frames ≈ 366 ms).

3.3.2 2. Spatio-Temporal Token Anchoring

Each vocabulary token w_k is permanently bound to a designated cortical neuron $n(w_k) \in \{1, \ldots, N\}$. Transition probabilities across rhetorical states are conditioned on the instantaneous postsynaptic membrane potential of target

somas:

$$P(w_{t+1} = j \mid w_t = i) \propto A_{ij} \cdot \left(\frac{v_{n(j)} - (-75)}{105} \right) \cdot \left(1 + 3 \cdot \mathbf{1}_{\{v_{n(j)} \geq 30\}} \right) \tag{11}$$

Neurons undergoing depolarization act as probabilistic attractors. Furthermore, when token j is emitted, it delivers a sharp $+28\,\text{mV}$ depolarizing current into $n(j)$, triggering a physical traveling wave packet across the axonal mesh.

4 Experimental Results & Video Verification

To evaluate the stability and emergent dynamics of the system, a full 60-second video sequence (3,600 frames at 60.00 FPS, 1280×840) was compiled and rendered on workstation node `bazina` (AMD GPU, Debian 13) and distributed to the unified storage node `boxC`.

The execution was choreographed across five distinct biophysical phases:

- **Act 1 (0–10 s): Awakening & Precession.** Spontaneous pacemaker bursting (5% intrinsic nodes) establishes baseline network rhythms ($\Phi \approx 42\,\text{Hz}$, cadence $\tau = 12\,\text{fps}$).
- **Act 2 (10–23 s): DC Depolarization Sweep.** A sensory electrode applies $+16\,\text{mV}$ across cortical layers. The global firing rate spikes to $\Phi \approx 135\,\text{Hz}$, accelerating the presidential speech tempo to $\tau = 6\,\text{fps}$. In-situ 3D semantic labels float above active somas (Fig. 2).
- **Act 3 (23–36 s): 40 Hz Gamma Entrainment.** Rhythmic gamma pulse stimulation entrains large-scale pyramidal synchronization. The vector scope reveals high-order closed Lissajous orbits.
- **Act 4 (36–50 s): Deep Volumetric Traversal.** Extreme 3D camera precession and zoom (1.4×). The coordinate manifold converges to MSE = 0.041, generating continuous tonic fields.
- **Act 5 (50–60 s): GABA Clamp & Phosphor Decay.** A hyperpolarizing clamp ($-22\,\text{mV}$) quenches somatic firing. Verbal cadence slows to $\tau = 22\,\text{fps}$ before settling into pure CRT analog phosphor decay.

5 Comparative Discussion

Pešek's twenty-year archival body anticipated three mainstream machine learning paradigms:

1. **Implicit Coordinate Representations:** `Neural32` (2009) demonstrated image reconstruction via continuous coordinate mapping over a decade before SIREN (Sitzmann et al., 2020) and NeRF (Mildenhall et al., 2020), utilizing CPU-based online backpropagation.
2. **Embodied Spiking Neuromorphics:** Pešek's 3D spatial topologies anticipated modern neuromorphic compute fabrics (e.g., Intel Loihi), recognizing that physical distance and axonal delays provide natural temporal basis functions without recurrent backpropagation through time.
3. **Grounded Autoregressive Sequence Generation:** Rather than decoupling tokens from substrate, Pešek's One-Button Presidential Speech engine provides a blueprint for biological grounding: language as a macroscopic acoustic manifestation of underlying cellular oscillations.

6 Conclusion

The computational lineage of Kryštov Pešek (Kof) represents a cohesive body of independent research uniting low-level systems engineering, biophysics, and generative semantics. By synthesizing his 2007–2026 discoveries into `kofsNeuralWorks`, we demonstrate that complex cybernetic intelligence can emerge without cloud dependencies or tensor frameworks. Future extensions will pipe real-time 50 Hz European electrical grid frequency micro-deviations from the `observator` platform into the sensory electrode array, completing the bridge between planetary telemetry and biophysical synthetic speech.

Code & Artifact Availability

The complete source code and rendered media of `kofsNeuralWorks` are publicly preserved at:

- Local Repository: `file:///home/kof/src/2026/10-rijen/kofsNeuralWorks/`
- Git Remote: `git@github.com:KOF/2026.git`
- Master Video (60 s): `/mnt/data/output/kofsNeuralWorks_full_minute.mp4`

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